



Multidimensional gene regulatory landscape of a bacterial pathogen in plants

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Understanding the gene regulation of plant pathogens is crucial for pest control and thus global food security. An integrated understanding of bacterial gene regulation in the host is dependent on multi-omic datasets, but these are largely lacking. Here, we simultaneously characterized the transcriptome and proteome of a bacterial pathogen in plants. We found a number of bacterial processes affected by plant immunity at the transcriptome and proteome levels. For instance, salicylic acid-mediated plant immunity suppressed the accumulation of proteins comprising the tip component of the bacterial type III secretion system. Interestingly, there were instances of concordant and discordant regulation of bacterial messenger RNAs and proteins. Gene co-expression analysis uncovered previously unknown gene regulatory modules underlying virulence. This study provides molecular insights into the multiple layers of gene regulation that contribute to bacterial growth in planta, and elucidates the role of plant immunity in affecting pathogen responses.

The growth of bacterial pathogens in plants and disease development are determined by genetically encoded bacterial virulence and plant immune systems¹. Despite the wealth of available knowledge concerning these systems in isolation, interactions between the two—especially how plant immunity affects bacterial function—are poorly understood². Previously, it was shown that in planta transcriptomics of the bacterial pathogen *Pseudomonas syringae* pv. *tomato* DC3000 (*Pto*) can be used to identify bacterial messenger RNAs whose expression is influenced by plant immune activation³. The study investigated the roles of two major forms of plant immunity, pattern-triggered immunity (PTI) and effector-triggered immunity (ETI), in affecting the transcriptomes of *Pto* in *Arabidopsis thaliana* leaves at an early stage of infection. ETI was triggered using *Pto* strains expressing effectors such as AvrRpt2 (ref. 4,5). Activation of PTI or ETI dramatically changed the transcriptomes of *Pto*, and the degree of transcriptome changes correlated well with bacterial growth in plants. Because the salicylic acid (SA) pathways are important components of ETI⁶, the impact of ETI on bacterial transcriptomes was strongly compromised in single and double mutants of *SID2* and *PAD4*, key components of the SA pathways, as well as the AvrRpt2 receptor mutant *rps2 rpm1*. Intriguingly, a bacterial iron-acquisition pathway was commonly suppressed by PTI and ETI. In planta transcriptome analysis has also been conducted in fungal pathogens. One study profiled co-transcriptomes of *A. thaliana* and a variety of *Botrytis cinerea* isolates to reveal gene co-expression networks of the pathogens that underlie virulence mechanisms⁷. Another study analysed transcriptomes of the fungal pathogen *Colletotrichum* at multiple stages of infection in *A. thaliana* and maize, revealing transcriptome signatures associated with the lifestyle transition of the fungal pathogens in planta⁸, highlighting that analysis of pathogen responses at different infection stages can also be informative.

Although transcriptome analysis is a useful and widely used approach for elucidation of cellular function, it has been well established that mRNA expression does not always reflect protein expression and thus it becomes clear that a better understanding of cellular behaviour requires direct interrogation of protein expression^{9,10}. A previous study showed that proteome analysis of leaf commensal bacteria can reveal metabolic changes in bacteria residing on the leaf surface¹¹. However, the capacity of proteomics to describe plant-associated bacteria remains limited¹². For instance, analysis of bacterial responses in the intercellular space (apoplast) of leaves, which is an important niche for various commensal and pathogenic bacteria, poses a major challenge because the large preponderance of plant material relative to bacterial material confounds analysis. To date, there is no proteome study of bacterial pathogens in the leaf apoplast and, thus, we lack comprehensive knowledge of bacterial proteins that are affected by host plants during infection. Moreover, due to the lack of comparative analyses spanning different modalities of bacterial responses in planta, little is known about the flow of bacterial genetic information (for example, from mRNA expression to protein expression) that is important for virulence and how this is affected by plant immunity.

Here, we simultaneously profiled bacterial transcriptomes and proteomes in planta and identified bacterial processes influenced by plant immunity at the mRNA and protein levels at early and late stages of infection. Comparative analysis of transcriptomes and proteomes revealed that changes in bacterial mRNA and protein expression are correlated in general. However, there are functions that are selectively changed at the mRNA or protein level. Furthermore, gene regulatory network analysis of bacteria showed previously unknown gene regulatory modules that mediate bacterial virulence in planta. Together, this study reveals the multi-layered regulatory mechanisms that underlie interactions between plants and bacterial pathogens.

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